

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : PCG-CS501 Compiler Design

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.
Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 × 10 = 10]

- (i) If all the operators are binary, then a string of operands and operators is a postfix expression if and only if _____
- (ii) Given a grammar $G=(V, T, P, S)$ and every production in P is of the form $A \rightarrow \alpha$ where A is in V and α is in $(V \cup T)^*$, then G is _____
- (iii) A compiler running on computers with a small memory would normally be _____
- (iv) Input to LEX is _____
- (v) A basic block is _____
- (vi) Given a finite automaton $M=(Q, \Sigma, \delta, q_0, F)$. If δ maps $Q \times \Sigma$ to 2^Q , then _____
- (vii) Given a string abc , the string acc is a _____
- (viii) A garbage is _____
- (ix) How many descriptors are used for track both the registers (for availability) and addresses (location of values) while generating the code?
- (x) A synthesized attribute is an attribute whose value at a parse tree node is defined in terms of _____
- (xi) Elimination of loop invariant computation is a peephole optimization. True/False?
- (xii) _____ is a loop optimization

Group-B (Short Answer Type Question)

Answer any three of the following

[5 × 3 = 15]

- 2. Describe input buffering in lexical analyser. [5]
- 3. Explain the model of a non recursive predictive parser with a diagram. [5]
- 4. Find the output, given grammar G_1 and associated semantic rules and input: aadbd [5]
 - $S \rightarrow AS \{ \text{print}(1) \}$
 - $S \rightarrow AB \{ \text{print}(2) \}$
 - $A \rightarrow a \{ \text{print}(3) \}$
 - $B \rightarrow bC \{ \text{print}(4) \}$
 - $B \rightarrow dB \{ \text{print}(5) \}$
 - $C \rightarrow c \{ \text{print}(6) \}$
- 5. What is ambiguity? Show that $G_2: \{ S \rightarrow aS | Sa | a \}$ is ambiguous [5]
- 6. What is code optimization? Optimize the following C-code: [5]

```
count=0;
result=0;
while(count++ < 20)
{
    increment= 2*count;
    result += increment;
}
```

Group-C (Long Answer Type Question)

Answer any three of the following

[15 × 3 = 45]

7. For the following grammar [3+5+5+2]
- $E \rightarrow E \text{ or } T \mid T$
 $T \rightarrow T \text{ and } F \mid F$
 $F \rightarrow \text{not } F \mid (E) \mid 0 \mid 1$
- Eliminate left recursion from the above grammar
 - Find $\text{FIRST}(X)$, $\text{Follow}(X)$ for each variable in the grammar
 - Construct a predictive parser table for the grammar
 - Is the above grammar LL(1). Justify your answer [2+10+3]
8. a) What is a compiler? [9+6]
- Explain the different phases of compiler with an example
 - Compare and contrast between a compiler and an interpreter
9. i) Express the expression $y = (a+b) * c$ in
- postfix notation
 - Abstract syntax tree
 - Three address code
- ii) Implement the TAC using
- quadruples [3+6+3+3]
 - triples
 - Indirect triples
10. Consider the regular expression $(a+b)^* a(a+b)(a+b)$
- Augment the expression and construct the syntax tree for the above regular expression
 - Find $\text{Firstpos}()$ and $\text{Lastpos}()$ for every internal node in the syntax tree
 - Find $\text{Followpos}()$ for every position in the syntax tree
 - Construct the corresponding DFA for the given RE using $\text{Followpos}()$ [1+5+9]
11. a) What is LEX?
- Explain the working of LEX
 - Show the step by step construction of a lexical analyzer with the following three tokens
 - a
 - abb
 - a*b+

*** END OF PAPER ***